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A Novel Error Metric for Evaluating the Error Correction Capability of Approximate Units

Yuin Dou, Haroon Waris, Roger Woods, *Senior Member, IEEE* and Weiqiang Liu, *Senior Member, IEEE*

Abstract—The use of Approximate Arithmetic Units (AAUs) in an error-tolerant application enhance circuit performance and reduce power consumption. Recently, a lot of technical work is reported on the Approximate High-Level Synthesis (AHLS) where these AAUs are used to create high-quality approximate system-level designs. To enhance the quality of approximate designs produced by AHLS the existing studies primarily focus on search algorithms within AHLS, while neglecting the fact that the optimal approximate designs exhibit a higher frequency of effective error capability. Therefore, the objective of this work is to delve into the error correction capabilities of AAUs that have not received adequate attention and integrate them in AHLS design. This effort provides new perspectives and methods to improve the quality of AHLS design exploration. To achieve this, we propose a novel metric for evaluating the error correction capabilities of AAUs, detailing its calculation method and properties. We applied this metric to two AHLS case studies and experimentally validated its effectiveness as AHLS achieved superior approximate designs.

Index Terms—Approximate computing, DSE, Number-aware Pruning, GA

I. INTRODUCTION

The approximate arithmetic units (AAUs) use an approximation design philosophy that allows controlled computational errors to significantly reduce energy consumption and enhance computation speed, optimizing computing resources while maintaining system performance. The AAUs have already been successfully applied in error-tolerant applications. For example, [1] replaced the precise operations of a Gaussian filter with an approximate multiplier, resulting in 32.5% power savings. Additionally, [2] described the design of an approximate multiplier for machine learning, which improved the energy efficiency by 1.7x (up to 2.1x) with negligible impact on machine learning accuracy.

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Corresponding author: Weiqiang Liu.

Y. Dou is with the College of Integrated Circuit Science and Engineering, Nanjing University of Posts Telecommunications (NJUPT), China, 210023. E-mail: douyuqin@njupt.edu.cn

H. Waris is with the Department of Electrical Engineering, Institute of Space Technology, Islamabad, Pakistan, E-mail: haroon.waris@ist.edu.pk

R. Woods is with the School of Electronics, Electrical Engineering and Computer Science, Queen's University Belfast, Belfast BT9 5AH, U.K. E-mail: r.woods@qub.ac.uk;

W. Liu is with College of Electronic and Information Engineering, Nanjing University of Aeronautics and Astronautics (NUAA), China, 211106. E-mail: liuweiqiang@nuaa.edu.cn

Recently, a new research focus has emerged on how to combine different AAUs to minimize errors and maximize the approximation benefits¹. The existing state-of-the-art approach to address this challenge is Approximate High-Level Synthesis (AHLS), which can automatically search for different combinations of AAUs and select an optimal combination for the final approximate design. The research on AHLS primarily focuses on improving outcomes through search algorithms or error estimation; however, it overlooks the error correction capabilities of AAUs [3] [4] [5]. The error correction capability of AAU refers to utilize its own inherent errors to correct input errors. The inclusion of this feature can improve the results of AHLS. In high-quality approximation designs, the AAUs tend to have a higher frequency of triggering error correction (Section V will introduce the relationship between error correction capabilities and optimal approximate design.).

To better understand and utilize this capability, we propose a new metric for evaluating the error correction capability of AAUs. Moreover, we have systematically devised a calculation method and properties of this metric. We applied the metric to two AHLS case studies and experimentally validated its effectiveness in helping AHLS achieve optimal designs. This work not only provides a new perspective for approximate design but also opens new avenues for optimizing efficient computing systems in AHLS.

II. BACKGROUND

A. Approximate arithmetic unit

The AAUs optimize energy consumption, area, and speed of the circuit on account of minimal accuracy loss. The current design methodologies for AAUs primarily include logic simplification and voltage scaling. The voltage scaling technology achieves power reduction by lowering the voltage of the circuit. However, this typically impacts the speed and stability of the circuit, and the control over circuit accuracy becomes challenging. Contrary, the logic simplification involves reducing the complexity and power consumption of the circuit by simplifying the computational logic, such as by merging certain logic gates or reducing the number of bits. The advantage of this method is that it allows for direct control over the source and magnitude of errors, facilitating precise adjustments to the performance and accuracy of the approximate circuit. Therefore, modern research in the design of AAUs predominantly focuses around the logic simplification. A large number of open-source high-quality AAUs are now available

¹Approximate benefits refer to the savings in area or power consumption achieved through the use of approximate units.

in the technical literature because in recent years researchers have given significant attention to this research domain.

For example, the EvoApprox8b [6] contains 500 designs of various AAUs, [7] has recently provided the designs for approximate adders and approximate multipliers. These approximate units entail different error characteristics, and there have been recent studies concerning the varying errors of AAUs. In, [8] the security of CNNs has been enhanced by utilizing the error variance of approximate units. In the domain of hardware security the intruders try to steal the information, the [9] demonstrated that the errors in approximate arithmetic units can be used to obscure information leakage. In the work of [10], it was first discovered that errors in approximate circuits could play a role in error correction, although the systematic analysis was not presented.

B. Approximate High-Level Synthesis

The emergence of numerous AAUs has greatly advanced the field of AHLS. The AHLS aims to utilize approximation techniques to automatically generate designs that achieve high approximation benefits while satisfying the error constraints. In recent studies, [4] has used approximate units as core technology and combined space pruning and genetic algorithms to improve the exploration quality of AHLS. [3] employed a method based on prior knowledge to enhance design quality while reducing the exploration time. [11] first constructed an approximate library using approximate units and then applied reinforcement learning to search for excellent approximate designs. [12] utilized a jump algorithm to effectively search the best designs and designed a metric to select approximate units to ensure outstanding results. However, the method for selecting approximate units in this work overlooked the error correction capability between approximate units.

In summary, these studies primarily focus on improving algorithms to achieve better approximate designs, but they neglect the fundamental reason for selecting excellent approximate designs—research on the error correction capability of approximate units. In this work, we proposed a systematic methodology that leverages AHLS to achieve better designs by incorporating the error compensation mechanism of AAUs.

III. MOTIVATION

The multiply-add circuit in Fig. 1 is analyzed as a motivational example, which is a commonly used structure in digital signal processing and machine learning. Nodes a_0 and a_1 in Fig. 1 are configured with the approximate multipliers Mul8u_EXZ and Mul8u_150Q, respectively, both sourced from the literature [6]. Fig. 1 also shows their corresponding mean error distance (MED) and area savings². The difference between Fig. 1(a) and Fig. 1(b) is that Fig. 1(b) uses an approximate adder for the addition operation. Generally, the use of AAUs brings greater area savings and power savings to the circuit. The area savings in Fig. 1(b) are also significantly greater than those in Fig. 1(a). Correspondingly, the error of

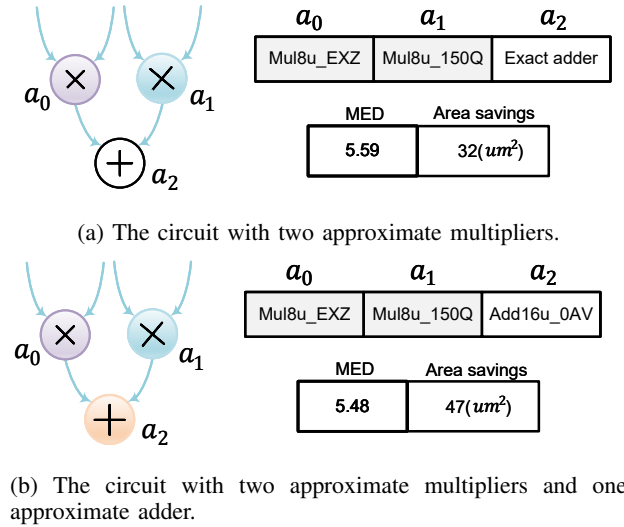


Fig. 1: The error and area savings of the multiply-accumulate circuit.

Fig. 1(b) would theoretically be worse than Fig. 1(a). However, the comparison results of the two figures show that the MED in Fig. 1(b) is smaller than that in Fig. 1(a). This indicates that the approximate adder Add16u_0AV has mitigated the can make the initial errors introduced by the two approximate multipliers in Fig. 1(a) smaller. This is an interesting phenomenon, as the introduction of another approximate units not only did not increase the error but actually reduced the initial overall errors.

We assume that the input error of the addition operation under the input i is $e_0^i + e_1^i$, where e_0^i is the error introduced by Mul8u_EXZ, and e_1^i is the error introduced by Mul8u_150Q. Thus, the sum of output errors in Fig. 1(a) is $sum1 = \sum_{i=1}^n e_0^i + e_1^i$, and the error in Fig. 1(b) is $sum2 = \sum_{i=1}^n e_0^i + e_1^i + e_2^i$, where e_2^i is the error introduced by Add16u_0AV. To ensure that $sum2$ is less than $sum1$, there must exist a significant number of e_2^i to meet $e_0^i + e_1^i + e_2^i < e_0^i + e_1^i$. The above analysis shows that it is precisely due to the error of add16u_0AV that the error of the approximate configuration in Figure 6.1(b) is reduced. The individual error of add16u_0AV acts as a correction for the overall system error, which also means that the approximate adder add16u_0AV is more suitable for this approximate configuration. Hence, we can conclude that in certain cases the individual errors in the AAUs can improve the initial errors of the system (in the above example, e_0 and e_1 can be regarded as the initial system error). This is a special error handling capability of AAUs, and in this work, we define this ability as AAU_{Ecc} . The example shown above is a motivation for this research work, and we are able to achieve high-quality results. Moreover, we believe this research will be a foundation for new research direction in the domain of AHLS.

IV. AAU_{Ecc}

A. AAU_{Ecc} Evaluation

The evaluation of AAU_{Ecc} require the analysis of related parameters. such as the initial errors of the system, which are

²The error estimates in Figure 6.1 use the same 100,000 random test vectors as inputs. Area savings refer to the difference between the area of the exact circuit and that of the approximate circuit.

the errors that the AAU is expected to improve. The expected improvement error (EIE) is defined as follows:

$$EIE = \sum abs \{E_1, E_2, E_3 \dots E_m\} \quad (1)$$

where $\{E_1, E_2, E_3 \dots E_m\}$ refers to all possible initial errors in the circuit. In this formula, $E_i = Out_i^{err} - Out_i^{exact}$, which represents the difference between the initial output and the exact result for all possible inputs. EIE denotes the sum of the absolute values of all E_i .

Secondly, the errors corrected by AAU need to be considered. This is also a key factor in evaluating the error correction capability of the approximate unit. The actual improvement error (AIE) is defined as:

$$AIE = \sum \{X_1, X_2, X_3 \dots X_n\} \quad (2)$$

where $X_i = |E_i| - |Out_i^{Axe} - Out_i^{exact}|$, and Out_i^{Axe} is the output using the AAU that needs to be evaluated

Similarly, the errors introduced to the system by the AAU are considered by using Increased error(INE):

$$INE = \sum \{Y_1, Y_2, Y_3 \dots Y_m\} \quad (3)$$

where $Y_i = |Out_i^{Axe} - Out_i^{exact}| - |E_i|$.

Finally, considering only errors is not sufficient because a small number of inputs cannot adequately evaluate the AAU_{Ecc} . Therefore, the total number of inputs and the frequency changes also need to be considered. Thus, we have defined the following metric:

$$AAU_{Ecc} = \frac{(AIE - INE) * (n + m)}{EIE * N} \quad (4)$$

where N is the number of all possible inputs, n is the number of X_i , and m is the number of Y_i . This metric clearly demonstrates the capability of AAU to handle input errors under the current inputs.

AAU_{Ecc} can only be used to evaluate the error correction capability of a single AAU. However, an approximate design may utilize multiple AAUs. When analyzing the overall phenomenon of error correction within an approximate design, the AAU_{Ecc} becomes invalid. Therefore, we define a new metric for the evaluation, which is calculated by summing the AAU_{Ecc} values of multiple approximate units and then taking the average.

$$Ave.AAU_{Ecc} = \frac{\sum_{k=1} AAU_{Ecc}k}{k} \quad (5)$$

V. PROPERTY ANALYSIS

A. The properties of AAU_{Ecc}

Table I is used as an example to analyze the properties of AAU_{Ecc} . In Table I, a_0 , a_1 , and a_2 represent the nodes of the circuit in Fig 1, where a_0 and a_1 are multiplication operations, and a_2 is an addition operation. a_2 is the AAU to be estimated, MED1 refers to the error of a_0 and a_1 , MED2 refers to the error of a_0 , a_1 and a_2 . The term 'utilization' refers to the ratio of area savings to the errors introduced, where a higher value is clearly preferable. In this table, MED1, MED2, and AAU are all calculated using the same 100,000 random test vectors.

In this table, we can observe that the initial error of the system is MED1 = 4.954. When AAU_{Ecc} is less than 0, the approximate adder of a_2 introduces errors into the system, as its MED2 is bigger than MED1. Furthermore, given the same initial error, the smaller the AAU_{Ecc} , the worse introduced error. Conversely, in the last row, when AAU_{Ecc} is greater than 0, the approximate adder of a_2 improves the errors of system, as MED2 is less than MED1. Moreover, we can see that the variation of 'utilization' aligns with AAU_{Ecc} . This can be explained as follows; a larger AAU_{Ecc} indicates that AAU have more errors that play a role in correcting the error of system. Consequently, with the same approximate benefits, the errors introduced into the system decrease. This can provide us with an insight while selecting an appropriate approximate adder from the set of available approximate adders to configure node a_2 . Typically, we would utilize the calculation method of "utilization" which involves calculating the error first and then performing complex physical synthesis. However, we can also use AAU_{Ecc} instead of "utilization" at this point, thereby avoiding the complex physical synthesis process.

In summary, we can conclude two properties of AAU_{Ecc} :

- The larger the value of AAU_{Ecc} , the smaller the error introduced, and the more errors are corrected. When AAU_{Ecc} is greater than 0, it indicates that the error characteristics of the AAU can improve the error of system.
- The value of AAU_{Ecc} also reflects the 'utilization' of the approximate unit. This implies that AAU_{Ecc} can be used for the preliminary selection of approximate units, allowing users to assess the 'utilization' without complex physical synthesis.

TABLE I: The analysis for the properties of AAU_{Ecc}

a0	a1	a2	utilization	MED1	AAU_{Ecc}	MED2
AxMul1	AxMul2	add16u_0P8	AxMul2	4.954	-0.24206	14.0456
AxMul1	AxMul3	add16u_0HE	AxMul3	4.954	-0.11721	5.598
AxMul1	AxMul3	Add16u_0AV	AxAdd1	5.598	1.19047	5.487

AxMul1: Mul8u_150Q, AxMul2: Mul8u_2P7, AxMul3:Mul8u_EXZ. All these approximate arithmetic units come from [6]

B. The properties of $Ave.AAU_{Ecc}$

The $Ave.AAU_{Ecc}$ reflects the average error correction capability of the approximate units in the approximate design. The Fig. 2 is used to analyze the properties of $Ave.AAU_{Ecc}$. Fig 2 illustrates the exploration process of ENAP in finding the optimal approximate design for 2x2 convolution under different constraints (MED<15 and MED<20). Each small black dot represents an approximate design, with the Y-axis showing the area savings of the approximate design and the X-axis showing the error of the approximate design. All the approximate designs shown in Fig. 2 meet the error constraints, but the area savings of each approximate design are different. Clearly, the greater the area savings, the better the approximate design. To analyze the characteristics of $Ave.AAU_{Ecc}$, we present the area savings and corresponding $Ave.AAU_{Ecc}$ of the approximate designs within the red dashed line in Table II.

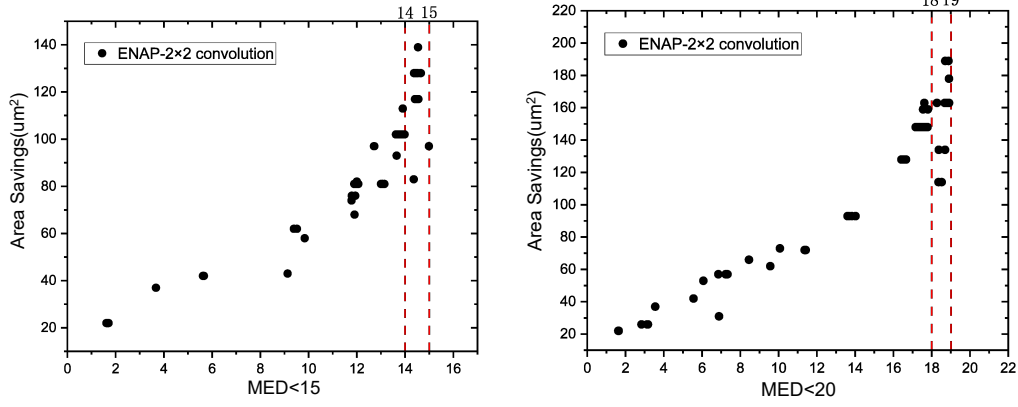


Fig. 2: The exploration process of 2x2 convolution

TABLE II: The properties of $Ave.AAU_{Ecc}$

MED<15		MED<20	
Area savings	$Ave.AAU_{Ecc}$	Area savings	$Ave.AAU_{Ecc}$
83	-0.30654	114	-0.11065
97	-0.23643	134	-0.06534
117	-0.11162	163	-0.05026
128	-0.08364	178	-0.03842
139	-0.05278	189	0.11106

From Table II, it can be observed that the $Ave.AAU_{Ecc}$ of the approximate designs increase with the area savings. This demonstrates that $Ave.AAU_{Ecc}$ effectively reflects the quality of the approximate design. This phenomenon can be easily explained. In approximate designs, a higher $Ave.AAU_{Ecc}$ indicates that more errors are being corrected. This provides a greater opportunity to use more AAUs or larger area-saving AAUs, consequently leading to greater area savings. Therefore, like AAU_{Ecc} , $Ave.AAU_{Ecc}$ can be used for the preliminary evaluation of approximate designs, eliminating the need for physical synthesis by the user.

VI. APPLICATION CASE

A. Formation of the initial configurations

In ENAP [4], genetic algorithm(GA) is employed as the search algorithm to obtain high-quality approximate designs. However, the results of GA are constrained by the initial designs. In the original framework, ENAP lacked suitable criteria for creating initial designs; thus, a random method was adopted to create these initial designs. In this work, AAU_{Ecc} is utilized to assist in the formation of initial designs, which is shown as follows:

Step1: Initially, a node is selected at random, and an AAU is configured randomly.

Step2: Starting from the node of the previously configured AAU, the next step involves sequentially selecting the AAU with the maximum AAU_{Ecc} from the approximation library.

Step3: Finally, the process concludes either when the configuration of all nodes is complete or when the error constraint is exceeded, thus forming an initial design.

The FIR filter is used as an application of this experiment, comprising of 24 nodes, including 13 multiplication operations and 12 addition operations. This work use ENAP framework to explore two methodologies. The difference lies in the method used to generate the initial design: the first method follows the aforementioned method, while the second method employs a random approach to form the initial design. Fig. 3 illustrates the search process and results of the two methods under different error constraints. The black dots represent the search process of the initial ENAP, while the red dots represent the search process of ENAP using AAU_{Ecc} , with both methods iterated 10 times. The legend shows the highest $Ave.AAU_{Ecc}$ of the initial designs for both methods, calculated according to Equation (5). As shown in Fig. 3, regardless of the error constraint, the search results of ENAP using AAU_{Ecc} are significantly better than those of the initial ENAP. Moreover, the search curve of the initial ENAP does not converge to a result after ten iterations, indicating that many good results have not been found. In contrast, ENAP using AAU_{Ecc} converges to a result, for example, when the $MED < 5$ and $MED < 15$, it begins to converge at the 7th iteration, and the results at the start of convergence are much better than those of the initial ENAP after ten iterations.

B. Selection for Approximate Units

[12] employed the jump search(JS) for the design space exploration of approximate circuits. This method initially configures the first node, and then sequentially configures subsequent nodes by selecting approximate units from an approximation library. The selection of these approximate units is governed by the FOM(Figure of Merit) defined in [12]. The FOM does not account for variations in error propagation; it only selects the approximate unit that provides the greatest area savings at that moment. Therefore, it does not guarantee the selection of the most suitable approximate unit. We modified the JS framework by adopting AAU_{Ecc} as the criterion for selecting approximate units and the approximate unit with the highest AAU_{Ecc} is selected for design each time.

The 2×2 convolution operation is used as an application for this experiment. It consists of 7 nodes, including 4 multiplication operations and 3 addition operations. The Fig. 4

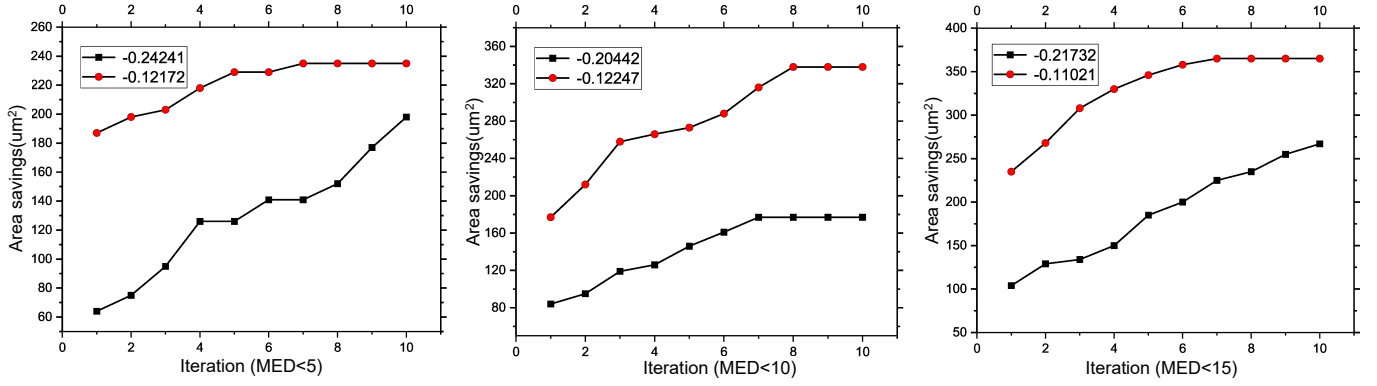


Fig. 3: The search results of ENAP using different methods to generate the initial design.

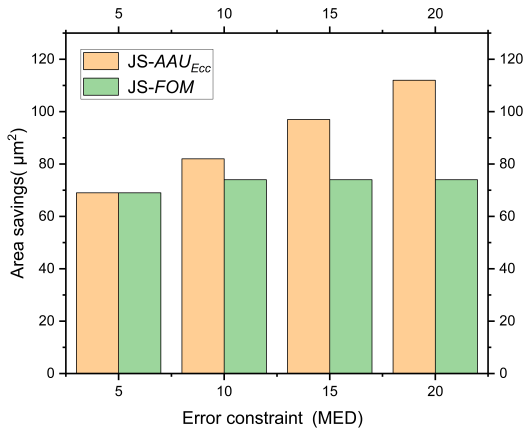


Fig. 4: The search results of JS using two metrics.

shows the search results using two different metrics. The Y-axis represents area savings; thus, a higher value indicates better quality of the approximate design. The X-axis represents different error constraints, using MED as the evaluation metric for error quality. The Fig. 4 demonstrates that search results using AAU_{Ecc} as the metric for selecting approximate units are better than FOM, especially when the error constraints increase. The primary difference between AAU_{Ecc} and FOM is that AAU_{Ecc} considers the current error characteristics when selecting approximate units. It selects high-fitness approximate units based on the current error, making the use of approximate units more valuable. In contrast, FOM does not consider the current error, often leading to identical search results under different error constraints. For instance, in Fig. 4, the search results for MED=10 and MED=20 are the same, despite the different error constraints.

VII. CONCLUSION

In this work, we have defined a novel error metric to assess the error correction capabilities of approximation units. Moreover, this metric has been applied to two case studies within the AHLs framework. The experimental results demonstrate that AHLs achieved better results by taking into consideration the

proposed metric. This work is the first to investigate the error correction capability of approximate units, aiming to improve the design exploration quality of AHLs and provide a new research perspective for AHLs.

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